

SoW: Developing a Physics-Informed Particle Impact Ignition Model for High-Pressure Oxidizing Environments – Year 4

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1. Objective/Requirements

The overall scope of the assessment is to develop an experimentally calibrated tool to predict *a priori* whether a given system is susceptible to catastrophic failures via particle impact (PI) based on expected operating conditions (gas temperature, pressure, concentration, flow rate), materials (composition, properties, particle features), and component geometry. This tool will accelerate design of new components, assessments of existing hardware, and dispositioning of anomalies. This past year (Year 3), we carried out multi-physics finite element simulations of particle impacts and gained mechanistic insight into key field variables across a range of environmental conditions. We also integrated the physics-based models into a quantitative predictive framework, the Particle Impact Ignition Risk Assessment Tool (PIIRAT), which can assess the risk of PI for idealized spherical particles in complex geometries under realistic operating environments. This SoW is for an additional one (1) year Phase 4, in which PIIRAT will be validated and refined through targeted sensitivity/validation testing and through the integration of additional key physics (transient and multiparticle effects).

MIT will support the assessment through the following tasks.

Task 1: Revise the particle ignition modeling, particle combustion modeling, and target ignition implementation to consider additional key physics. These modifications address limitations in the current embodiment and are essential to modeling accurately the tube-bend validation tests under **Task 2**.

- MIT has implemented models of the three (3) phases of PII – particle ignition, particle combustion, and target kindling. Each model will be modified to account for oxygen concentration effects, which can strongly influence behaviors. The particle ignition lookup table will be re-generated with oxygen concentration treated as an independent parameter, using an oxidation law appropriate for oxygen-dependent kinetics. Target ignition modeling will account for oxygen-concentration-dependent particle flame temperature when computing heat flux from a combusting particle to the target surface. The particle combustion model will treat oxygen concentration as an independent parameter in determining particle flame temperature and overall combustion time.
- The **particle ignition model** will be modified to consider deformable target materials. Past work has assumed a rigid sapphire target, providing overly conservative estimates of particle ignition velocity, thus limiting model validity. Here we will develop look-up tables of particle ignition conditions specific to the material combinations tested (e.g., Ti on Ti impacts). The modeling workflow will be documented for subsequent extension to additional material pairs, based on NASA interest.

- The **combustion time model** will be modified to consider spatial variations in gas temperature and pressure. This modification is essential for predicting accurately the trajectories of burning particles.
- The **target kindling model** will be modified to consider multi-particle effects.
- MIT will analyze experimental results generated by WSTF, compare results to relevant models, and adjust modeling parameters to calibrate, repeating until modeling results represent experimental findings. Success will be demonstrated when calibrated model outputs reproduce key experimental observables (ignition outcome/probability, combustion time, target ignition outcomes).
- Each model (including debris transport model) will be modified to account for transient flow solutions. The PIIRAT sub-models will interface with the framework described in the open literature to capture transient flow conditions and/or time-varying geometry. Success will be demonstrated by conducting analysis on a transient CFD simulation, showing particle paths interacting with transient flow fields and/or time-varying geometry and sub-models sampling environment conditions as a function of time.

Task 2: Extend quantitative risk assessment framework to real geometries and develop general engineering guidelines.

- Update existing workflow with PIIRAT model refinements and apply to three exemplary components – pipe bends, a stationary manual valve and the first stage of an oxygen-rich turbine
- Comparison to in-situ high-speed video of pipe-bend experiments.
- Develop general guidelines informing ignition risk as a function of geometric parameters and operating conditions.
- A sensitivity analysis will be carried out to determine how statistical distributions in particle size, particle velocity, and incidence angle affect the probability of ignition.

2. Characteristics, Scope, and Specs

MIT shall perform the following duties:

- Provide general technical input during all team meetings
- Provide input on particle impact failure modes and root causes
- Review available test data
- Provide input on test materials and methods to anchor models
- Develop predictive models
- Review and provide inputs to assessment documentation including the Stakeholder Briefing and Final Report.
- Visit test facility(ies) to support test design and verification (maximum of one trip each per year to either WSTF, LaRC, or MSFC)
- Participate in weekly team meetings (1-2 hours per week)
- Participate in Technical Interchange Meetings (1-2 days)
- Participate in Stakeholder Briefing (1-2 hours)

3. Restrictions on Publishing

MIT can publish data related to this assessment with the exception of proprietary data related to proprietary particle impact data and information that is restricted by International Traffic in Arms Regulations (ITAR) and/or Export Administration Regulations (EAR). All documentation (e.g. manuscripts, presentations, etc.) resulting from this effort shall be reviewed (within 60 days of submission) by Greg Harrigan, Stephen Peralta, and Jonathan Tylka, and approved for the presence of confidential or proprietary material prior to publication.

4. Place of Performance

Massachusetts Institute of Technology (MIT gas turbine laboratory, MIT Institute for Solider Nanotechnologies, Cordero Lab)

5. Period of Performance

Work shall be performed from authority to proceed through March 2027.

6. Schedule

Task	Description	Start	End
1.1	Revision of unit models		
1.1.1	Revised particle ignition model (deformable targets)	2/25/2026	4/15/2026
1.1.2	Revised particle combustion (varying O2 pressure)	4/15/2026	5/15/2026
1.1.3	Revised target kindling (multi-particle effects)	5/15/2026	6/15/2026
2.1	PIIRAT simulations of real-world geometries		
2.1.1	Tube-bend simulations	2/25/2026	8/15/2026
2.1.2	Manual valve simulations	6/15/2026	10/15/2026
2.1.3	Turbine stage simulations	6/15/2026	2/15/2027

7. Project Milestones and Deliverables

#	Milestone Description	Deliverables	Date
1	Technical Interchange Meeting #4 (WSTF)	Overview of model status	6/16/2026
2	Mid-point Review (MIT)	Review MIT/WSTF test progress	8/15/2026
3	Complete Task 1.1	Reporting on updated unit models	9/15/2026
4	Complete Task 2.1	Report on analysis of PIIRAT sims	1/10/2027
5	Technical Interchange Meeting #5 (MIT)	Report on experimental results/ integrated model/recommendations for transition strategies	12/2/2026
6	Annual report	1. Stakeholder Briefing Slides 2. Annual report inputs	2/20/2027

Test Data Reports shall include test conditions, configurations, and key results

Mid-point Review Slides shall be provided in powerpoint format per the NESC Assessment team template (to be provided by NASA). The slides shall include an update on progress of the models and usage plan and shall provide an overview of the accomplishments to-date, a description of remaining activities, and an estimated progress at completion of the effort.

Model drafts shall be described and/or demonstrated to the assessment team via Microsoft Teams in such a format as to be understood and reviewed by the NESC Assessment Team.

A *Usage Plan draft* shall be sent, in Microsoft Word format, to all NESC Assessment Team members for review.

Final Models shall be provided to the NESC Assessment Team for review and will be considered “Approved” with concurrence from Greg Harrigan, Stephen Peralta, Jonathan Tylka, and Michael Fischels.

The *Stakeholder Briefing* is the final presentation for the work conducted under this assessment. An overview of the models and usage plan shall be provided in powerpoint per the NESC Stakeholder Briefing template (to be provided by NASA) for inclusion in the compiled Stakeholder Briefing package.

The *NESC Final Report* is a word document intended to capture the details of an NESC Assessment. Final Reports Inputs shall be provided, in Microsoft Word per the NESC Final Report template (to be provided by NASA). Specific content of the report is to be determined during the assessment activity. At a minimum, inputs shall include details of the approach taken for each model, the results of the model, and a detailed description of model capabilities.